3:

Solution:

1. A is the union of open intervals $\left(-1,-\frac{1}{2}\right)\cup\left(\frac{1}{2},1\right)$ so A is open in $\mathbb{R}$ since $A\subseteq Y$, A is open in Y
2. $B=\left[-1,-\frac{1}{2}\right)\cup\left(\frac{1}{2},1\right]$ is open in Y since $B=Y\cap\left(\left(-2,-\frac{1}{2}\right)\cup\left(\frac{1}{2},2\right)\right)$ which is an open set of $\mathbb{R}$. B is not open in $\mathbb{R}$ since $1+\epsilon\notin B\,\forall\epsilon>0$.
3. $C=\left(-1,-\frac{1}{2}\right]\cup\left[\frac{1}{2},1\right)$ is not open in Y or $\mathbb{R}$ since $\frac{1}{2}-\epsilon<\frac{1}{2}\Rightarrow\frac{1}{2}-\epsilon\notin C$
4. D is clearly not open in Y or $\mathbb{R}$
5. E is open in Y and $\mathbb{R}$ since the density of $\mathbb{Q}$ in $\mathbb{R}$ ensures $\exists\,\epsilon>0$ s.t. $(y-\epsilon,y+\epsilon)\subseteq Y$ for $y\in Y$.

4:

Solution:

Proof. Every open set in $X\times Y$ can be represented as a union of basis elements that are open rectangles of the form $U_i\times V_i$ for $U_i\in X$ open and $V_i\in Y$ open. By definintion we have

$$\pi_1(U\times V)=\{\pi_1(u\times v)\mid(u,v)\in U\times V\}=\{u\mid(u,v)\in U\times V\text{ for some }V\}$$

Now if $V=\varnothing$ then $U\times V=\varnothing$ and hence $\pi_1(U\times V)$ is open in X. If $V\neq\varnothing$ then by definiton $\pi_1(U\times V)$ is open in X since for every $u\in U$ and any $v_0\in V$ the pair $(u,v_0)\in U\times V$ so $u\in\pi_1(U\times V)$. Also, every element of $\pi_1(U\times V)$ is some u with $(u,v)\in U\times V$ so that $u\in U$ and $\pi_1(U\times V)=U$ which is open in X. A near identical argument exists for $\pi_2:X\times Y\to Y$. □

6:

Solution: It is clear that

$$\beta=\{(a,b)\times(c,d)\mid a,b,c,d,\in\mathbb{Q}(a<b,c<d)\}$$

covers $\mathbb{R}^2$ since any $(x,y)\in\mathbb{R}^2$ will be in β, even for $x,y\in\mathbb{R}\setminus\mathbb{Q}$ due to the density of $\mathbb{Q}$ in $\mathbb{R}$. It is also clear that β is countable since we may define the map $(a,b)\times(c,d)\in\beta\mapsto(a,b,c,d)\in\mathbb{Q}^4$. Now, for some $(x,y)\in B_1\cap B_2$ with $B_1,B_2\in\beta,\exists\,B_3\in\beta$ s.t. $(x,y)\in B_3$. However, since elements of β are open rectangles in $\mathbb{R}^2$, $B_1\cap B_2$ is also an open rectangle and hence we take $B_3=B_1\cap B_2$ so $(x,y)\in B_3\subseteq B_1\cap B_2\in\beta$. Thus β is a basis.

We now show β generates the standard topology on $\mathbb{R}^2$. Let β' denote the standard basis of open rectangles $(a,b)\times(c,d)$ with $a,b,c,d\in\mathbb{R}$. We show β and β' generate the same topology by mutual refinement.

(β refines into β', trivially): Since $\mathbb{Q}\subseteq\mathbb{R}$, every $B\in\beta$ is already an element of β', so for any $B\in\beta$ and $x\in B$, take B itself as the element of β' contained in B.

(β' refines into β): Let $B'=(a,b)\times(c,d)\in\beta'$ with $a,b,c,d\in\mathbb{R}$, and let $(x,y)\in B'$, so $a<x<b$ and $c<y<d$. By density of $\mathbb{Q}$ in $\mathbb{R}$, choose $a',b',c',d'\in\mathbb{Q}$ with

$$a<a'<x<b'<b,\qquad c<c'<y<d'<d.$$

Then $B=(a',b')\times(c',d')\in\beta$ and $(x,y)\in B\subseteq B'$.

Since every element of β' can be refined by an element of β containing any given point, and vice versa, β and β' generate the same topology. Hence $\mathcal{T}_\beta$ is the standard topology on $\mathbb{R}^2$.

8:

Solution: The set L must be in one of the following forms:

$$L_1 = \{(x,n) \mid x \in \mathbb{R}, n \in \mathbb{R} \text{ constant}\} \quad \text{(horizontal line)}$$
$$L_2 = \{(n,y) \mid y \in \mathbb{R}, n \in \mathbb{R} \text{ constant}\} \quad \text{(vertical line)}$$
$$L_3 = \{(x,y) \mid y = mx + b, m, b \in \mathbb{R} \text{ constant}\} \quad \text{(line with slope } m\text{)}$$

i.e. either a horizontal line, vertical line, or a line with linear slope. $\mathbb{R}_\ell \times \mathbb{R} = [a,b) \times (c,d)$ for $a,b,c,d \in \mathbb{R}$ so $L_1 \cap (\mathbb{R}_\ell \times \mathbb{R})$ is either empty if $n < b$ or $n > d$ or

$$L_1 \cap (\mathbb{R}_\ell \times \mathbb{R}) = \{[x,n) \mid x \in \mathbb{R}(a \leq x < c)\}.$$

Similarly, we have

$$L_2 \cap (\mathbb{R}_\ell \times \mathbb{R}) = \{(n,y) \mid y \in \mathbb{R}(b < y < d)\}.$$

if $a < n < c$, and empty otherwise. Finally

$$L_3 \cap (\mathbb{R}_\ell \times \mathbb{R}) = \{[x,y) \mid x,y \in \mathbb{R}(a \leq x < c, b < y < d)\}$$

and hence L_1 inherits the lower-limit topology along with L_3 while L_2 inherits the subspace topology. Now, consider $\mathbb{R}_\ell \times \mathbb{R}_\ell = [a,b] \times [c,d]$. Once again

$$L_1 \cap (\mathbb{R}_\ell \times \mathbb{R}_\ell) = \{[x,n) \mid x \in \mathbb{R}(b < n < d)\}$$
$$L_2 \cap (\mathbb{R}_\ell \times \mathbb{R}_\ell) = \{[n,y) \mid y \in \mathbb{R}(a \leq n \leq c)\}.$$

However, we now split L_3 into $L_{3'}$ and L_4 where $L_{3'}$ is L_3 with $m \geq 0$ and L_4 is L_3 with $m < 0$. Then

$$L_{3'} \cap (\mathbb{R}_\ell \times \mathbb{R}_\ell) = \{(x,y) \mid x \in [p,q), y \in [c,d)\}$$
$$L_4 \cap (\mathbb{R}_\ell \times \mathbb{R}_\ell) = \{(x,y) \mid x \in (p,q], y \in [c,c)\}$$

where p,q are the left and right endpoints of the intersection of L with $\mathbb{R}_\ell \times \mathbb{R}_\ell$. Hence $L_1, L_2, L_{3'}$ all have the lower limit topology, while L_4 has the upper limit topology.

9:

Solution: Since $\mathbb{R}_d$ is the discrete topology and contains all singletons, the basic open products are

$$\{x\} \times V, x \in \mathbb{R} \text{ and } V \text{ open in } \mathbb{R}.$$

The other topology has basis elements consisting of open intervals $((x_1,y_1),(x_2,y_2))$.
Case $1(x_1 = x_2 = x_0)$: In this case we have

$$\{(x_0,y) \mid y_1 < y < y_2\} = \{x_0\} \times (y_1,y_2)$$

which is a basis element of $\mathbb{R}_d \times \mathbb{R}$ since (y_1,y_2) is open in $\mathbb{R}$.
Case $2(x_1 < x_2)$: Now, consider when $x = x_1, x_1 < x < x_2$, and $x = x_2$. One $x = x_1$ points (x_1,y) satisfy $(x_1,y_1) < (x_1,y)$ so $y > y_1$ and since $x_1 < x_2$ ther eis no bound from $y_2 : \{x_1\} \times (y_1,\infty)$. For $x_1 < x < x_2$ any (x,y) has $(x_1,y_1) < (x,y) < (x_2,y_2) : \{x\} \times \mathbb{R}$. Now if $x = x_2$, similar to before we have $\{x_2\} \times (-\infty,y_2)$.
Putting it all together we see

$$((x_1,y_1),(x_2,y_2)) = (\{x_1\} \times (y_1,\infty)) \cup \left(\bigcup_{x_1 < x < x_2} \{x\} \times \mathbb{R} \right) \cup (\{x_2\} \times (-\infty,y_2))$$

So that every part is of the form $\{x\} \times V$ for V open in $\mathbb{R}$ making them basis elements.
Now, every product basis element $\{x\} \times V$ (for V open in $\mathbb{R}$, a union of open intervals (y_1, y_2)) is a dictionary-order basic interval:

$$\{x\} \times (y_1, y_2) = ((x_1, y_1), (x_2, y_2))$$

which we showed is a basis element earlier.
This $\mathcal{T}_{\text{dict}} = \mathcal{T}_{\mathbb{R}_d \times \mathbb{R}}$ and is finer than the standard topology on $\mathbb{R}^2$ since singletons $\{x\} \times V$ are open here and not in $\mathbb{R}^2$.

3:

Solution:

Proof. Assume A, B are sets such that $A \subseteq X$ and $B \subseteq Y$ are closed. It follows that $A \times B \subseteq X \times Y$ is closed. This is because its complement $(A \times B)^c$ is open since

$$\begin{aligned}(A \times B)^c &= (X \times Y) \setminus (A \times B) \\ &= (X \times (X \setminus A)) \times (Y \times (Y \setminus B))\end{aligned}$$

but $X \setminus A$ is open and so is $Y \setminus B$. □

11:

Solution:

Proof. Let X, Y be Hausdorff. Consider $(x_1, y_2), (x_2, y_2) \in X \times Y$. Since X and Y are Hausdorff it follows that there exists neighborhoods $U_1, U_2 \subseteq X$ and $V_1, V_2 \subseteq Y$ s.t. $U_1 \cap U_2 = \varnothing$ and $V_1 \cap V_2 = \varnothing$ where $x_1 \in U_1, x_2 \in U_2, y_1 \in V_1, y_2 \in V_2$. Hence $(x_1, y_1) \in U_1 \times V_1 \subseteq X \times Y$ and $(x_2, y_2) \in U_2 \times V_1 \subseteq X \times Y$ and $(U_1 \times V_1) \cap (U_2 \times V_2) = \varnothing$ and thus $X \times Y$ is Hausdorff. □

13:

Solution:

Proof. ($\Rightarrow$) Assume X is Hausdorff and consider $(x, y) \in X \setminus \Delta$ so that $x \neq y$. Then there exists open sets u, V such that $(x, y) \in U \times V \subseteq (X \times X) \setminus \Delta$ and that $x \in U, y \in V$, and $U \cap V = \varnothing$. It follows that no $(z, z) \in \Delta$ and $(z, z) \in U \times V$ since then $z \in U$ and $z \in V$ which is a contradiction so $(X \times X) \setminus \Delta$ is open and thus Δ is closed in $X \times X$.
($\Leftarrow$) Now take Δ to be closed in $X \times X$ so that $(X \times X) \setminus \Delta$ is open. Thus $\exists U, V$ open such that $U \times V$ open with $(x, y) \in U \times V$ for $x \neq y$ and $U \times V \subseteq (X \times X) \setminus \Delta$. Since $(U \times V) \cap \Delta = \varnothing$ it follows that $x \in U$ and $y \in V$ and that $U \cap V = \varnothing$ since otherwise the intersection with Δ would be nonempty. Thus, X is Hausdorff. □

14:

Solution: $(x_n) = \frac{1}{n}$ converges to all $x \in \mathbb{R}$ since every x_n and x_m (with $m \neq n$) are distinct so at most a finite amount of them are contained in the finite set $R \setminus U$ so infinitely many line in U: past some $n > N$, every $x_n \in U$.

19:

Solution:

(a) Suppose $x \in \operatorname{Int} A$ so $\exists U$ open with $x \in U \subseteq A$ so $U \cap (X \setminus A) = \varnothing$. Thus U is an open set containing x but disjoint from $X \setminus A$ so $x \notin \overline{(X \setminus A)}$ and hance $x \notin \operatorname{Bd} A$ and $\operatorname{Bd} A \cap \operatorname{Int} A = \varnothing$. Now $\operatorname{Int} A \subseteq A \subseteq \overline{A}$ and $\operatorname{Bd} A \subseteq \overline{A}$ so $\operatorname{Int} A \cup \operatorname{Bd} A \subseteq \overline{A}$. Now take $x \in \overline{A}$. If $x \in \operatorname{Int} A$ we are done. If $x \notin \operatorname{Int} A$ then no open set containing x lies in A. Equivalently $\forall U$ open such that $x \in U, U \not\subseteq A$ and hence $U \cap (X \setminus A) \neq \varnothing$ so $x \in \overline{(X \setminus A)}$ so since $x \in \overline{A}, x \in \overline{A} \cap \overline{(X \setminus A)} = \operatorname{Bd} A$

(b) ($\Leftarrow$) Suppose A is clopen, so $A = \overline{A}$ and $X \setminus A$ closed and thus $X \setminus A = \overline{X \setminus A}$. It follows that

$$\operatorname{Bd} A = A \cap (X \setminus A) = \varnothing$$

($\Rightarrow$) Let $\operatorname{Bd} A = \varnothing$. By (a),

$$\overline{A} = \operatorname{Int} A \cup \operatorname{Bd} A = \operatorname{Int} A$$

But $\operatorname{Int} A \subseteq A \subseteq \overline{A}$ so

$$\operatorname{Int} A \subseteq A \subseteq \overline{A} = \operatorname{Int} A$$

forcing $A = \operatorname{Int} A = \overline{A}$ so A is clopen.

(c) ($\Rightarrow$) Let U open so $\operatorname{Int} U = U$. By (a) $\overline{U} = \operatorname{Int} U \cup \operatorname{Bd} U = U \cup \operatorname{Bd} U$ which is a disjoint union and so $\overline{U} \setminus U = \operatorname{Bd} U$.
($\Leftarrow$) Suppose $\operatorname{Bd} U = \overline{U} \setminus U$ so $\operatorname{Bd} U \cap U = \varnothing$. Let $x \in U$ so $x \in \overline{U}$ and by (a) $\overline{U} = \operatorname{Int} U \cup \operatorname{Bd} U$. If $x \notin \operatorname{Int} U$ then $x \in \operatorname{Bd} U$ which is a contradiction since $\operatorname{Bd} U \cap U = \varnothing$ and thus $x \in \operatorname{Int} U$. Hence $U \subseteq \operatorname{Int} U$ so $U = \operatorname{Int} U$.

(d) No, let $U = (0,1) \cup (1,2)$ in $\mathbb{R}$ so $\overline{U} = [0,2]$ and $\operatorname{Int} \overline{U} = (0,2)$ but $1 \in \operatorname{Int} \overline{U}$ and $1 \notin U$.

[0] LaTeX code for this document can be found on github here